SANKALP PANDEY

Computer Science Scholar · Software Engineering & Robotics Intern Candidate

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EDUCATION

Newton School of Technology, Ajeenkya DY Patil University — Pune, India — **Expected 2030**

Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence & Machine Learning) — **First Year Scholar**

Core Coursework: Data Structures & Algorithms, Systems Programming, Object-Oriented Design (C++), Linear Algebra.

TECHNICAL SKILLS

Languages:	Python, C, C++, JavaScript, TypeScript, Rust (exploring)
Frontend & Web:	React, Next.js (App Router), HTML5, CSS3, Tailwind CSS
Backend & APIs:	FastAPI, Django, REST Route Handlers, Microservices
Databases & DevOps:	PostgreSQL, Prisma ORM, Supabase, Docker, Linux (Bash), Vercel, Git/GitHub
Robotics & Hardware:	Arduino, Microcontrollers (PWM, I2C, SPI), Sensor Integration, ROS (Robot Operating System fundamentals)

FEATURED TECHNICAL PROJECTS

Beyond 2D — Deterministic 2D-to-3D Architectural Vision Engine [Live Studio | GitHub] — **Python, FastAPI, OpenCV, Three.js, Gemini AI**

- Engineered a deterministic vision engine converting static 2D blueprint drawings into watertight, photorealistic 3D models in <3s.
- Eliminated a critical 21-room mesh fragmentation bug by implementing morphological closing passes and room polygon filters in OpenCV.
- Developed parametric door and window placement logic, preventing exterior opening hallucinations by validating contours against Gemini AI.
- Built mobile-first interactive touch navigation featuring sliding drawer sheets, a virtual D-pad, and pinch-zoom in a Three.js viewport.

CampusHub — Verified College Student Community Platform [Live App | GitHub] — **Next.js, TypeScript, PostgreSQL, Prisma, Android**

- Architected a full-stack campus platform with student feeds, peer-to-peer marketplace, event registries, and direct messaging threads.
- Designed and normalized a 24-model relational database schema in Prisma backed by PostgreSQL with role-based access control.
- Implemented institutional email allowlisting with cryptographic OTP verification to guarantee 100% verified campus membership.
- Packaged the Next.js application into a native Android APK using Capacitor, consuming a shared REST API backend.

NOVA — Multi-Tenant Workspace with Inline Contextual AI [Live App | GitHub] — **Next.js, TypeScript, PostgreSQL, Prisma, Radix UI**

- Built an agile productivity workspace with projects, tasks, and notes backed by JWT-authenticated session security.
- Integrated contextual inline AI drafting and summarization into the editor canvas using sentence-buffered streaming commits.
- Engineered keyboard-first command navigation and optimistic UI updates for instantaneous task state mutations.

OPEN-SOURCE CONTRIBUTIONS (AUDITED)

Supabase — *Isomorphic JavaScript/TypeScript Client (supabase/supabase-js)* — **PR #2802 · Open**

- Authored a bug fix preventing broadcast payloads containing an `ids` key from crashing the Realtime client's filter transformation.

Kestra — *Declarative Orchestration & Workflow Engine (kestra-io/blueprints)* — **PR #4870 · Merged**

- Engineered an automated orchestration blueprint scanning Python `requirements.txt` dependencies for vulnerabilities against PyPI.

Squidbrake — *Safety & Change-Control Guard for Autonomous AI Agents* — **PR #186, #184, #183, #151 · Merged**

- Implemented automated CLI safety rules classifying database drop commands, system service alterations, and audit log purges as irreversible.
- Added network guard rule detection identifying suspicious DNS lookups engineered to exfiltrate private system data.

ROBOTICS EXPERIENCE & HACKATHONS

School Robotics Club — *Core Competitor & Team Lead* — **3-Year Full Tenure**

- Designed, wired, and programmed autonomous mobile robots utilizing microcontrollers, ultrasonic/infrared sensors, and motor drivers.
- Led pit crews and hardware tuning during competitive matches, winning multiple interschool robotics championships.
- Calibrated PID control loops for line-following speed trials and obstacle-avoidance arenas under real-time match constraints.

Hackathons & Competitive Sprints — **2024 – Present**

- **Smart India Hackathon (SIH):** Collaborated in a national sprint delivering rapid prototype systems for public problem statements.
- **Hacktoberfest:** Successfully authored and landed 15+ verified pull requests across international open-source repositories.
- **ScratchStorm & Upcoming:** Competed in competitive team prototypes; currently building Project Orbit for Construct 2.0.